MDTS4214_737_8

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PROBLEM SET 8

1. Problem to demonstrate the fitting of a binary response variable using logistic regression

a. Consider the Weekly data available in the ISLR package of R.

Use the data from 1990 to 2008 as train data and the data on 2009-2010 as test data.

```r
#Package Required:
library(ISLR)

# Original Data Set-Weekly:
original_data=Weekly

# Knowing the Dataset:
# ??Weekly

# First few observations of the data set:
head(original_data)
```

```
##   Year   Lag1   Lag2   Lag3   Lag4   Lag5    Volume  Today Direction
## 1 1990  0.816  1.572 -3.936 -0.229 -3.484 0.1549760 -0.270      Down
## 2 1990 -0.270  0.816  1.572 -3.936 -0.229 0.1485740 -2.576      Down
## 3 1990 -2.576 -0.270  0.816  1.572 -3.936 0.1598375  3.514        Up
## 4 1990  3.514 -2.576 -0.270  0.816  1.572 0.1616300  0.712        Up
## 5 1990  0.712  3.514 -2.576 -0.270  0.816 0.1537280  1.178        Up
## 6 1990  1.178  0.712  3.514 -2.576 -0.270 0.1544440 -1.372      Down
```

Information on the Data set:

Name: Weekly S&P Stock Market Data.

Description: Weekly percentage returns for the S&P 500 stock index between 1990 and 2010.

Format:

A data frame with 1089 observations on the following 9 variables.

*(i)Year:*The year that the observation was recorded

*(ii)Lag1:*Percentage return for previous week

*(iii)Lag2:*Percentage return for 2 weeks previous

*(iv)Lag3:*Percentage return for 3 weeks previous

*(v)Lag4:*Percentage return for 4 weeks previous

*(vi)Lag5:*Percentage return for 5 weeks previous

*(vii)Volume:*Volume of shares traded (average number of daily shares traded in billions)

*(viii)Today:*Percentage return for this week

*(ix)Direction:*A factor with levels Down and Up indicating whether the market had a positive or negative return on a given week

Train-Test Split:

```
# Training Data [1990 to 2008]-
train_data=subset(original_data,Year>=1990 & Year<=2008)

# Test Data [2008 to 2010]-
test_data=subset(original_data,Year>=2009 & Year<=2010)
```

b. Using the training data, fit a logistic regression with Direction as the response and the five lag variables plus Volume as predictors.

Are there any significant predictors? If not, which of the tests is reporting the least p value.

Interpret the coefficient corresponding to that test.

Fitted Logistic Model:

```
#Logistic model-
logit_model=glm(Direction~Lag1+Lag2+Lag3+Lag4+Lag5+Volume,family=binomial,data=train_data);logit_model
```

```
##
## Call:  glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##     Volume, family = binomial, data = train_data)
##
## Coefficients:
## (Intercept)         Lag1         Lag2         Lag3         Lag4         Lag5
##     0.33258     -0.06231      0.04468     -0.01546     -0.03111     -0.03775
##      Volume
##    -0.08972
##
## Degrees of Freedom: 984 Total (i.e. Null);  978 Residual
## Null Deviance:       1355
## Residual Deviance: 1342  AIC: 1356
```

Model Summary to check the existence of significant predictors:

```r
summary(logit_model)
```

```
##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##     Volume, family = binomial, data = train_data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.33258    0.09421   3.530 0.000415 ***
## Lag1        -0.06231    0.02935  -2.123 0.033762 *
## Lag2         0.04468    0.02982   1.499 0.134002
## Lag3        -0.01546    0.02948  -0.524 0.599933
## Lag4        -0.03111    0.02924  -1.064 0.287241
## Lag5        -0.03775    0.02924  -1.291 0.196774
## Volume      -0.08972    0.05410  -1.658 0.097240 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1354.7  on 984  degrees of freedom
## Residual deviance: 1342.3  on 978  degrees of freedom
## AIC: 1356.3
##
## Number of Fisher Scoring iterations: 4
```

Interpretation: *(i)*Except 'Lag1', all the predictors are insignificant, have p-values greater that 0.05.

*(ii)*Test on Coefficient of Lag1 reports the least p-value of 0.033762.

*(iii)*The regression coefficient corresponding to Lag1 is -0.06231, the coefficient of Lag1 is negative, which indicates that an increase in Lag1 is associated with a decrease in the log-odds of the market going "Up". Specifically, for a one-unit increase in Lag1, the log-odds of the market moving up decreases by 0.06.

c. Estimate the odds of market going "up" at median values of the predictors.

```r
# the median values of predictors
medians <- apply(train_data[, c("Lag1","Lag2","Lag3","Lag4","Lag5","Volume")],
                 2, median)

# New data with median values
new_data <- data.frame(t(medians))

# Prediction of probability
prob <- predict(logit_model, newdata = new_data, type = "response")

# Computation of odds
odds <- prob / (1 - prob);odds
```

```
##        1
## 1.267475
```

e. Using the fitted model and the predictor values from the test data, find an optimal cut point that minimizes TPR(1-FPR).

Report the confusion matrix corresponding to the optimal cut-point value.

From the confusion matrix compute the test error rate.

```r
# Step 1: Predict probabilities
prob_test <- predict(logit_model, newdata = test_data, type = "response")

# Step 2: Use unique probabilities as cut-points
cutpoints <- sort(unique(prob_test))

results <- data.frame(cut = cutpoints, TPR = NA, FPR = NA, score = NA)

for(i in 1:length(cutpoints)) {

  pred <- ifelse(prob_test > cutpoints[i], "Up", "Down")

  cm <- table(Predicted = pred, Actual = test_data$Direction)

  # Extract values safely
  TP <- ifelse("Up" %in% rownames(cm) & "Up" %in% colnames(cm), cm["Up","Up"], 0)
  TN <- ifelse("Down" %in% rownames(cm) & "Down" %in% colnames(cm), cm["Down","Down"], 0)
  FP <- ifelse("Up" %in% rownames(cm) & "Down" %in% colnames(cm), cm["Up","Down"], 0)
  FN <- ifelse("Down" %in% rownames(cm) & "Up" %in% colnames(cm), cm["Down","Up"], 0)

  # Avoid division by zero
  TPR <- ifelse((TP + FN) == 0, 0, TP / (TP + FN))
  FPR <- ifelse((FP + TN) == 0, 0, FP / (FP + TN))

  results$TPR[i] <- TPR
  results$FPR[i] <- FPR
  results$score[i] <- TPR * (1 - FPR)
}

# Step 3: Optimal cut-point
best <- results[which.max(results$score), ]
best
```

```
##          cut      TPR       FPR     score
## 49 0.4556775 0.557377 0.4883721 0.2851697
```

Final Confusion Matrix:

```r
best_cut <- best$cut

final_pred <- ifelse(prob_test > best_cut, "Up", "Down")

conf_matrix <- table(Predicted = final_pred, Actual = test_data$Direction)

conf_matrix
```

```
##          Actual
## Predicted Down Up
##      Down   22 27
##      Up     21 34
```

Test Error Rate(Probablity of Type 1 Error):0.4883721

f. Is there any change in the test error rate if we have worked with only the two predictors lag 1 and lag 2?

```r
# Fit model with only Lag1 and Lag2
model_lag12 <- glm(Direction ~ Lag1 + Lag2,
                   data = train_data, family = binomial)
#Prediction on test data
prob_test_lag12 <- predict(model_lag12, newdata = test_data, type = "response")

#Find Optimal Cut Point
cutpoints <- sort(unique(prob_test_lag12))

results <- data.frame(cut = cutpoints, TPR = NA, FPR = NA, score = NA)

for(i in 1:length(cutpoints)) {

  pred <- ifelse(prob_test_lag12 > cutpoints[i], "Up", "Down")

  cm <- table(Predicted = pred, Actual = test_data$Direction)

  TP <- ifelse("Up" %in% rownames(cm) & "Up" %in% colnames(cm), cm["Up","Up"], 0)
  TN <- ifelse("Down" %in% rownames(cm) & "Down" %in% colnames(cm), cm["Down","Down"], 0)
  FP <- ifelse("Up" %in% rownames(cm) & "Down" %in% colnames(cm), cm["Up","Down"], 0)
  FN <- ifelse("Down" %in% rownames(cm) & "Up" %in% colnames(cm), cm["Down","Up"], 0)

  TPR <- ifelse((TP + FN) == 0, 0, TP / (TP + FN))
  FPR <- ifelse((FP + TN) == 0, 0, FP / (FP + TN))

  results$TPR[i] <- TPR
  results$FPR[i] <- FPR
  results$score[i] <- TPR * (1 - FPR)
}

best_lag12 <- results[which.max(results$score), ]
best_lag12
```

```
##          cut       TPR       FPR     score
## 32 0.5268024 0.7868852 0.5581395 0.3476935
```

```r
#Confusion Matrix
best_cut_lag12 <- best_lag12$cut

final_pred_lag12 <- ifelse(prob_test_lag12 > best_cut_lag12, "Up", "Down")
```

```r
conf_matrix_lag12 <- table(Predicted = final_pred_lag12, Actual = test_data$Direction)

conf_matrix_lag12
```

```
##          Actual
## Predicted Down Up
##      Down   19 13
##      Up     24 48
```

Test Error (Probability of Type 1 Error):0.5581395

Comment:

Reduction in the predictors have resulted in increase of test error rate.